

Analysis of the Use of Mobile Application to Advance Agricultural Traceability

Reffatti, Leonardo; Porto, Josiane Brietzke; Barbosa, Jorge Luis Victoria (2022.0)

ABSTRACT ORIGINAL

Context: Nova Aliança has been industrializing the production of grapes by 410 families of winegrowers in Serra Gaúcha, since 2011. Grapes, the basis for beverages, are traced from the producer to the industry. Problem: In Brazil, traceability of plant products is a requirement of the market and became mandatory with Joint Normative Instruction ANVISA/SDA No. 2 of 07/02/2018, aiming at monitoring pesticide residues. However, little is known about the potential of digital tools for agricultural traceability. Solution: This article presents results of the transformation of physical documents to digital of the production areas and the use of harvest lots to improve the tracking of fruit loads, through a mobile application, developed under Design Science Research. IS Theory: It deals with challenges of Information Systems applied to Agriculture, under the Design Theory and Design Science paradigm. Method: The method is characterized by observational, analytical, and exploratory technological research, operationalized by Design Science Research, with initial evaluation of the artifact via action research in this real context of use and quantitative analysis of the results. Summary of Results: The mobile application was used by 127 families, representing 30.97% in the 20/21 harvest. It replaced the records in physical documents completely, made it possible to improve the traceability of production in 20 million kilograms of grapes and minimized problems regarding cargo compliance. Contributions and Impacts in the IS Area: This solution has been widely used by farmers and the results point to improved traceability and reinforce viability for agriculture. © 2022 ACM.